

Stochastic Processes

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Welcome to My Book

These lecture notes accompany the Stochastic Processes course.

1 Introduction to Stochastic Processes

Learning Outcomes

After studying this chapter, students should be able to:

Remember

1. **Define** a stochastic process and its basic components.
2. **State** the meaning of the index set, state space, and sample path.

Understand

3. **Explain** the difference between a random variable and a stochastic process.
4. **Describe** how a stochastic process models a quantity that evolves over time under uncertainty.
5. **Classify** a stochastic process according to its time domain and state space.

Apply

6. **Formulate** a stochastic process for a real-world problem by defining the random variable, time domain, and state space.
7. **Identify** appropriate stochastic process models for simple actuarial and financial applications.

The course will cover the probabilistic framework for stochastic models of real-world applications with emphasis on actuarial work. We will illustrate some practical actuarial problems for which we will develop mathematical models, tools and techniques for analysing and quantifying the uncertainty of the problems.

1.1 Why Stochastic Processes?

Many quantities encountered in actuarial science, finance, engineering, and the natural sciences evolve over time. Some evolve in a predictable manner, whereas others are affected by chance. A stochastic process provides a mathematical framework for describing such random phenomena.

Consider the following questions.

- How many insurance claims will have occurred by time t ?
- What No-Claim Discount (NCD) level will a policyholder be assigned at the next policy renewal?
- Will a policyholder aged 70 survive to age $70 + t$?

- What will be the price of a stock one week from today?

Although these questions arise in different fields, they have two common features:

1. The quantity of interest changes over time.
2. The future value cannot be predicted with certainty.

Consequently, we require mathematical models that describe both the evolution of the quantity over time and the uncertainty associated with its future values. Such models are called **stochastic processes**.

Throughout this text, we will study several important stochastic processes, including:

- the No-Claim Discount (NCD) model,
- the Poisson process for modelling claim arrivals,
- Markov processes for modelling transitions between states.

Each of these processes has different mathematical properties, but all can be viewed as collections of random variables indexed by time.

Before introducing the formal definition, we first distinguish between a random variable and a stochastic process.

1.2 Random Variables versus Stochastic Processes

A **random variable** represents the outcome of a single random experiment. For example,

- the total number of claims reported today,
- the lifetime of a policyholder,
- the amount of a single insurance claim.

A random variable is typically denoted by

$$X.$$

Its value is unknown before the experiment is performed, but once the outcome is observed, the random variable takes a single value.

In contrast, many real-world quantities are observed repeatedly over time. Instead of a single random variable, we consider a sequence or collection of random variables, one for each point in time.

For example, let

$$X_n = \text{number of claims reported on day } n,$$

where

$$n = 0, 1, 2, \dots$$

Rather than studying each random variable separately, we study the entire collection

$$X_n : n = 0, 1, 2, \dots$$

Similarly, if time is measured continuously, we may write

$$X_t : t \geq 0,$$

where t can take any non-negative real value.

A stochastic process therefore consists of a family of random variables indexed by time. At each time point, there is a random variable describing the state of the system.

For example:

- In the No-Claim Discount (NCD) model, X_n denotes the NCD level assigned to a policyholder at the n th policy renewal.
- In a Poisson process, $N(t)$ denotes the total number of claims that have occurred by time t .
- In a survival model, X_t indicates whether a policyholder is alive or dead at time t .

These examples illustrate that stochastic processes differ according to

- how time is measured (discrete or continuous), and
- the possible values that the process can take (discrete or continuous).

The next section introduces these classifications before presenting the formal definition of a stochastic process.

1.3 Motivating Examples

The following examples illustrate a variety of stochastic processes that arise naturally in actuarial science. They will be studied in greater detail in later chapters.

Example 1.1 (No-Claim Discount (NCD) System). A **No-Claim Discount (NCD) system** is widely used in motor insurance to adjust a policyholder's premium according to their recent claims experience. A policyholder who remains claim-free is rewarded with a premium discount, whereas one or more claims may result in a reduction of the discount or an increase in premium.

An example of a seven-level NCD system is shown below.

Level	7	6	5	4	3	2	1
Premium	100%	75%	65%	55%	45%	40%	33%

Level 1 corresponds to the highest discount (lowest premium), whereas Level 7 corresponds to no discount.

A policyholder moves between levels according to the following rules.

- **Claim-free year:** Move down one level, with Level 1 remaining unchanged.
- **Levels 4–7**
 - One claim: move up one level (or remain at Level 7).
 - Two or more claims: move to Level 7.
- **Levels 2–3**
 - One claim: move up two levels.
 - Two claims: move up four levels.
 - Three or more claims: move to Level 7.
- **Level 1**
 - One claim: move to Level 4.
 - Two claims: move to Level 6.
 - Three or more claims: move to Level 7.

Typical actuarial questions include:

1. What is the expected number of policyholders in each NCD level after a given number of years?
2. What is the probability that a policyholder currently at a particular level will be at Level 7 after two years?
3. What is the probability distribution of the NCD level after five years?
4. What proportion of policyholders is expected to occupy each level in the long run?

To study these questions, let

X_n = the NCD level of a policyholder after the n th policy year,

where $n = 0, 1, 2, \dots$. The collection

$$X_n : n = 0, 1, 2, \dots$$

describes the evolution of the policyholder's NCD level over time.

In this example,

- time is measured in policy years and is therefore **discrete**;
- the state space is

$$S = 1, 2, \dots, 7,$$

which is also **discrete**.

Hence, the NCD system can be represented by a **discrete-time, discrete-state stochastic process**. In a later chapter, we will show that, under suitable assumptions, this process can be modelled as a discrete-time Markov chain.

Example 1.2 (Poisson Process). Suppose that claims arise from a portfolio of health insurance policies. Let

$$N(t) = \text{the total number of claims that have occurred by time } t,$$

where $t \geq 0$. Assume that claims occur randomly over time at an average rate of λ claims per unit time (e.g., per day or per week).

Typical actuarial questions include:

1. Suppose that, on average, 20 claims occur each day. What is the probability that more than 100 claims occur during a week?
2. What is the expected time until the next claim?
3. What is the probability that exactly 15 claims occur during the next day?
4. What is the probability that no claims occur during the next hour?

The collection

$$N(t) : t \geq 0$$

describes the evolution of the cumulative number of claims over time.

In this example,

- time is measured continuously and is therefore **continuous**;
- the state space is

$$S = 0, 1, 2, \dots,$$

which is **discrete**.

Hence, this is an example of a **continuous-time, discrete-state stochastic process**.

The stochastic process $N(t) : t \geq 0$ is called a **Poisson process**. It is one of the most widely used counting processes in probability and actuarial science. Although claims occur at a constant average rate, the exact times at which they occur are random.

Later, we will study several equivalent ways of describing a Poisson process. One approach focuses on the counting process $N(t) : t \geq 0$, while another focuses on the times between successive claims. Both viewpoints are useful and lead to many important properties of the Poisson process.

Example 1.3 (Survival Process). Life insurers are interested in modelling the future lifetime of policyholders in order to price life insurance products and calculate reserves. One of the primary objectives is to estimate the probability that a life survives for a specified period.

Suppose that we observe a sample of n independent lives, all aged between x and $x + 1$. For each life i , we record the following information:

- $x + a_i$: the age at which observation begins, where $0 \leq a_i < 1$;
- $x + b_i$: the age at which observation ends if the life is still alive, where $0 \leq b_i < 1$;

- $x + t_i$: the age at which observation ends because of either death or censoring;
- d_i : the event indicator, where

$$d_i = \begin{cases} 1, & \text{if life } i \text{ dies during the observation period,} \\ 0, & \text{if life } i \text{ is censored.} \end{cases}$$

For example, the following table contains observations on eight lives aged between 70 and 71.

Life	a_i	b_i	d_i	t_i
1	0.0	1.0	1	0.25
2	0.0	1.0	1	0.75
3	0.0	1.0	0	1.00
4	0.1	0.6	1	0.50
5	0.2	0.7	1	0.60
6	0.2	0.4	0	0.40
7	0.5	1.0	1	0.75
8	0.5	0.75	0	0.75

A natural actuarial question is:

How can we estimate the probability that a life aged 70 dies before age $70 + t$, or survives to at least age $70 + t$, for $t \in [0, 1)$?

To describe the survival status of a life over time, define

$$X_t = \begin{cases} 1, & \text{if the life is alive at age } x + t, \\ 0, & \text{if the life has died before age } x + t, \end{cases} \quad t \geq 0.$$

The collection

$$X_t : t \geq 0$$

describes the evolution of the life's survival status over time.

In this example,

- time is measured continuously and is therefore **continuous**;
- the state space is

$$S = 0, 1,$$

which is **discrete**.

Hence, this is an example of a **continuous-time, discrete-state stochastic process**.

The probability of principal interest is

$$P(X_t = 1 \mid X_0 = 1),$$

which represents the probability that a life alive at age x survives to age $x + t$.

Later in this course, we will show that, under suitable assumptions, this stochastic process can be modelled as a **continuous-time Markov process**. This model provides a powerful framework for estimating survival probabilities and analysing more general multi-state actuarial models, such as disability and long-term care insurance.

1.4 Formal Definition of a Stochastic Process

After studying several motivating examples, we are now ready to give a formal definition of a stochastic process.

A **stochastic process** is a collection of random variables

$$X_t : t \in T,$$

defined on a common probability space. Each random variable X_t describes the state of the system at time t , while the collection $X_t : t \in T$ describes how the system evolves over time.

Key Idea

A stochastic process is **not** a single random variable. It is a **collection of random variables**, where each random variable describes the state of the system at a particular time.

The notation consists of three essential components:

- **Index set (T):** The set of time points at which the process is observed. The index set is often referred to as the **time domain**. Depending on the application,
 - $T = 0, 1, 2, \dots$ represents **discrete time**, while
 - $T = [0, \infty)$ represents **continuous time**.
- **State space (S):** The set of all possible values that the random variable X_t can take. The state space depends on the application. For example,
 - the NCD process has a finite set of discount levels,
 - the Poisson process has state space $0, 1, 2, \dots$,
 - Brownian motion has state space $\mathbb{R}$.

⚠ Common Misconception

The **index set** and the **state space** describe different aspects of a stochastic process.

- The **index set** specifies **when** the process is observed.
- The **state space** specifies **what values** the process can take.

For example, in the Poisson process,

$$T = [0, \infty), \quad S = \{0, 1, 2, \dots\},$$

so the time domain is continuous while the state space is discrete.

- **Random variables (X_t):** For each time point $t \in T$, X_t is a random variable describing the state of the system at time t .

It is important to note that the random variables in a stochastic process are generally **dependent** and need not have identical probability distributions.

1.5 Components of a Stochastic Process

The formal definition of a stochastic process contains several important components. Understanding the role of each component provides a clearer picture of how stochastic processes are constructed.

1.5.1 Probability space

Every stochastic process is defined on a common probability space

$$(\Omega, \mathcal{F}, P),$$

where

- Ω is the sample space,
- $\mathcal{F}$ is the collection of events,
- P is the probability measure.

The probability space represents all possible outcomes of the underlying random experiment.

1.5.2 Index set

The **index set** T specifies the time points at which the process is observed.

Examples include

- $T = 0, 1, 2, \dots$ for yearly observations in an NCD system,
- $T = [0, \infty)$ for continuously observed claim arrivals,

- $T = [0, \infty)$ for monitoring the survival status of an individual over time.

The index set determines whether the process evolves in **discrete time** or **continuous time**.

1.5.3 State space

The **state space** S is the set of all possible values that the process may assume.

For the motivating examples,

Process	State space
NCD process	Finite set of NCD levels
Poisson process	$0, 1, 2, \dots$
Survival process	$0, 1$ (or equivalently Alive, Dead)

The state space may be **discrete** or **continuous**, depending on the application.

1.5.4 Sample paths (realizations)

Fix an outcome $\omega \in \Omega$. The values

$$X_t(\omega), \quad t \in T,$$

form a **sample path** (or **realization**) of the stochastic process.

A sample path describes one possible evolution of the system through time. Different outcomes in the sample space generally produce different sample paths.

For example,

- in the NCD process, one sample path records the sequence of NCD levels experienced by a particular policyholder;
- in the Poisson process, one sample path records the cumulative number of claims observed over time;
- in the survival process, one sample path indicates when an individual changes from the Alive state to the Dead state.

Since the underlying experiment is random, infinitely many sample paths are possible, each occurring with an associated probability.

Interpreting a Sample Path

A sample path is obtained by fixing one outcome of the underlying random experiment and observing how the process evolves over time. Different outcomes generally produce different sample paths.

1.6 Classification of Stochastic Processes

Two fundamental characteristics determine the type of a stochastic process:

- the **index set (time domain)**, which may be **discrete** or **continuous**; and
- the **state space**, which may also be **discrete** or **continuous**.

These two characteristics naturally classify stochastic processes into four categories, as shown in Table 1.1.

Table 1.1: Classification of stochastic processes according to their time domain and state space.

Time domain	Discrete state space	Continuous state space
Discrete time	Discrete-time Markov chains (e.g., NCD system)	Random walks, autoregressive models
Continuous time	Poisson processes, birth-death processes, continuous-time Markov chains	Brownian motion, diffusion processes

! Key Observation

Every stochastic process is classified according to **two independent characteristics**:

1. the type of **time domain** (discrete or continuous), and
2. the type of **state space** (discrete or continuous).

Together, these determine which of the four classes the process belongs to.

The three motivating examples introduced earlier can now be placed within this classification:

- The **No-Claim Discount (NCD) system** is a **discrete-time, discrete-state** stochastic process.
- The **Poisson process** is a **continuous-time, discrete-state** stochastic process.
- The **Survival process** is also a **continuous-time, discrete-state** stochastic process.

We now briefly discuss each of the four classes.

1.6.1 Discrete-Time, Discrete-State Processes

In this class, observations are made only at discrete time points, and the process can occupy only a finite or countably infinite number of states.

For example, in the No-Claim Discount (NCD) system,

$$X_n, \quad n = 0, 1, 2, \dots,$$

denotes the discount level after the n th policy year. If the insurer uses three discount levels,

$$S = 0, 1, 2,$$

then both the time domain and the state space are discrete.

Many actuarial models, including discrete-time Markov chains, belong to this class.

1.6.2 Continuous-Time, Discrete-State Processes

In this class, the process evolves continuously through time, but the possible states remain discrete.

The Poisson process introduced earlier is an important example. If

$$N(t)$$

denotes the cumulative number of claims reported by time t , then

$$T = [0, \infty), \quad S = 0, 1, 2, \dots$$

Similarly, the survival process

$$X_t \in \text{Alive, Dead}$$

has a continuous time domain but only two possible states.

Many important actuarial models, including Poisson processes, birth-death processes, and continuous-time Markov chains, belong to this category.

1.6.3 Continuous-Time, Continuous-State Processes

In these processes, both time and the state space are continuous.

For example, let

$$S_t$$

denote the total amount of claims accumulated by time t . Since claim amounts are measured on a continuous scale,

$$T = [0, \infty), \quad S = \mathbb{R}_{\geq 0}.$$

Brownian motion and diffusion processes are important examples of continuous-time, continuous-state stochastic processes. These models are widely used in finance, risk analysis, and credibility theory.

1.6.4 Discrete-Time, Continuous-State Processes

Sometimes a quantity evolves continuously but is observed only at discrete time points.

For example, suppose the accumulated claim amount is recorded only at the end of each day, week, or month. Then the observations form the sequence

$$S_0, S_1, S_2, \dots,$$

where the time domain is discrete, while the state space remains continuous.

Processes of this type commonly arise when observations are collected periodically or when continuous monitoring is impractical or expensive. Time series models such as autoregressive (AR) models belong to this class.

Modelling Considerations

The choice of state space often depends on the level of detail required in a model.

For example, claim amounts are naturally continuous quantities and may therefore be modelled using a continuous state space. However, if claim amounts are recorded only to the nearest baht (or another fixed monetary unit), the process may instead be approximated by a discrete-state model.

Similarly, although many real-world phenomena evolve continuously through time, they are often observed only at discrete time points. In such cases, a discrete-time stochastic process may provide an appropriate and more practical model.

Modelling Choice

The classification of a stochastic process depends on the mathematical model rather than the physical system itself.

For example, claim amounts may be modelled using a **continuous** state space or approximated by a **discrete** state space if they are recorded to the nearest monetary unit. Likewise, a continuously evolving quantity may be observed only at discrete time points.

1.7 Roadmap of the Course

The classification of stochastic processes presented in Table 1.1 provides a useful roadmap for the topics covered in these lecture notes. Following the Institute and Faculty of Actuaries (IFoA) CS2 syllabus (Chapters 1–5), the focus will be on stochastic processes with **discrete state spaces**, including both discrete-time and continuous-time models.

The main topics covered are:

- **Discrete-time Markov chains**, which model systems observed at discrete time points with transitions between a finite or countably infinite number of states.
- **Continuous-time Markov processes**, where transitions between states may occur at any point in time.

- **Poisson processes**, an important class of continuous-time Markov processes used to model random event occurrences.
- **Time-homogeneous** and **time-inhomogeneous** Markov models, which distinguish whether the transition probabilities (or transition rates) remain constant over time or vary with time.

By the end of these lecture notes, you will understand how these stochastic processes are constructed, analysed, and applied to actuarial modelling.